

## Spinal Cord Injury

Guideline Title: Spinal Cord Injury

Guideline Number: GUID-3203-T012

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| Revision dates: 6/8/2021                                                                     | Developed by: Air Care Team |
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| Signature:  | Signature:                  |
| Department Numbers: 3203                                                                     |                             |

CXXII. **PURPOSE:** To recognize the patient with a suspected or known spinal cord injury and utilize all resources to prevent any further injury or deficit to the patient.

CXXIII. **PATHOPHYSIOLOGY:**

All trauma patients must be assumed to have a spinal cord injury until proven otherwise, especially traffic accidents and falls. Injury may involve the bony skeleton, the cord itself or both. There may or may not be a neurological deficit.

Hyperextension, hyperflexion, axial loading, or rotational forces may cause blunt spinal injury.

Ligamentous injuries (rupture or tearing of ligaments) can cause subluxation of a vertebral body over another, increasing the risk of spinal cord damage. Spinous process fractures are painful but are considered stable fractures. Fractures of intervertebral articulations may be associated with subluxation, and comminuted fractures of the vertebral body may produce bony fragments that penetrate the spinal canal and have potential to damage the cord. Herniated tissue from disc rupture may result in nerve damage and/or cord compression.

Damage to the spinal cord at or about the level of the fifth cervical vertebral may result in paralysis of the diaphragm that may compromise the ability to ventilate. In elderly patients, occult cervical fractures may occur with minimal trauma and be difficult to detect on physical exam. Neurogenic shock may accompany a low cervical or high thoracic spine injury and due to a loss of sympathetic tone below the level of the injury hypotension and a paradoxical bradycardia occur, resulting in inadequate perfusion of tissues. The patient may have symptoms of autonomic nerve dysfunction evidenced by acute urinary retention and priapism. The patient in neurogenic shock loses the ability to regulate body temperature and may become hypo or hyperthermic.

CXXIV. **DEFINITIONS:**

A. MAP: Mean Arterial Pressure

CXXV. **DEPARTMENT GUIDELINE:**

A. Assess and manage airway, breathing, and circulation.

- i. The patient with a high cervical cord injury may breathe poorly and may require airway suction or intubation. Respiratory mechanical support may be needed using rapid-sequence intubation with in-line spinal immobilization per *GUID-G008 - Rapid Sequence Intubation (RSI) for Endotracheal (ET) Intubation*.

- B. Initiate full spinal precautions and maintain c-spine immobilization throughout the transport.
  - i. Immobilization of the neck and body should be maintained using cervical collar, straps, tape, and blocks.
- C. If a helmet is present:
  - i. Motorcycle helmet: should always be removed while maintain c-spine precautions as best as possible.
  - ii. If athletic headgear is present:
    - 1. If patient is protecting his airway and able to talk, headgear and shoulder pads should not be removed in the field. Remove the facial mask with assistance of athletic trainer.
    - 2. A football helmet and chin strap should only be removed if:
      - a. The helmet and chin strap do not hold the head securely, such that immobilization of the helmet does not also immobilize the head
      - b. The design of the helmet and chin strap is such that even after removal of the face mask the airway cannot be controlled, or ventilation be provided
      - c. The face mask cannot be removed after a reasonable period of time
      - d. The helmet prevents immobilization for transportation in an appropriate position
    - 3. If a football helmet is removed, remove helmet while simultaneously removing shoulder pads. (Cut string at the center of pads and cut the straps under the arms.)
- D. For interfacility transports where the immobilization devices have been removed, review radiology exams completed and then consult with sending or receiving physician regarding selective spinal motion restriction.
  - i. Suspected cervical spine injuries should always remain in a c-collar unless body habitus or clinical situation does not allow it.
  - ii. Suspected thoracic or lumbar spine injuries do not require a long back board but transfer to and from stretchers should always be performed with multiple staff members and a slide board to maintain spinal precautions.
- E. Patients may lack vasomotor control & cannot sweat below the lesion. Protect from hypothermia.
- F. Treat nausea/vomiting per *GUID-G004 - Motion Sickness/Nausea*.
- G. Treat pain and/or anxiety per *GUID-G005 – Analgesia and Sedation Management*.
- H. Assess for neurologic deficits (flaccid paralysis, loss of sensory, autonomic, or reflex function) and approximate level of injury.
- I. Assess for neurogenic shock: Signs include hypotension, warm extremities, normal or decreased heart rate, and wide pulse pressure. If patient displays signs of inadequate perfusion and all other causes except neurogenic shock are ruled out, resuscitation per *GUID-M011 – Hypotension* with preference to peripheral vasoconstrictors (i.e., phenylephrine or Norepinephrine). Recommend maintaining MAP >85-90.
- J. Assess for soft tissue injury, swelling, bony crepitus, pain, deformity, muscle spasm, movement, sensation, strength of extremities, abdominal breathing, use of accessory muscles, body temperature, lack of bowel control, priapism.
- K. Monitor for bradycardia. Bradycardia may require external pacing or administration of atropine per *GUID-C004 - Bradycardia*. This complication usually occurs in severe, high cervical (C1 through C5) lesions.

CXXVII.

## REFERENCES

- A. Bilello, J.F., Davis, J.W., Cunningham, M.A., et al. (2003). Cervical spinal cord injury and the need for cardiovascular intervention. *Arch Surg*, 138: 1127.
- B. Hadley, M. N, et al. (2002). Blood pressure management after acute spinal cord injury. *Neurosurgery*, 50:S58.
- C. Jia, X., Kowalski, R.G., Sciubba, D.M., Geocadin, R.G. (2013). Critical care of traumatic spinal cord injury. *J Intensive Care Med*, 28: 12.
- D. Theodore, N., Aarabi, B., Dhall, S.S., et al. (2013). Transportation of patients with acute traumatic cervical spine injuries. *Neurosurgery*, 72(Suppl2):35-39.

CXXVIII.

## APPENDIX

- A. Sensory Impairment Related to Level of Spinal Injury

